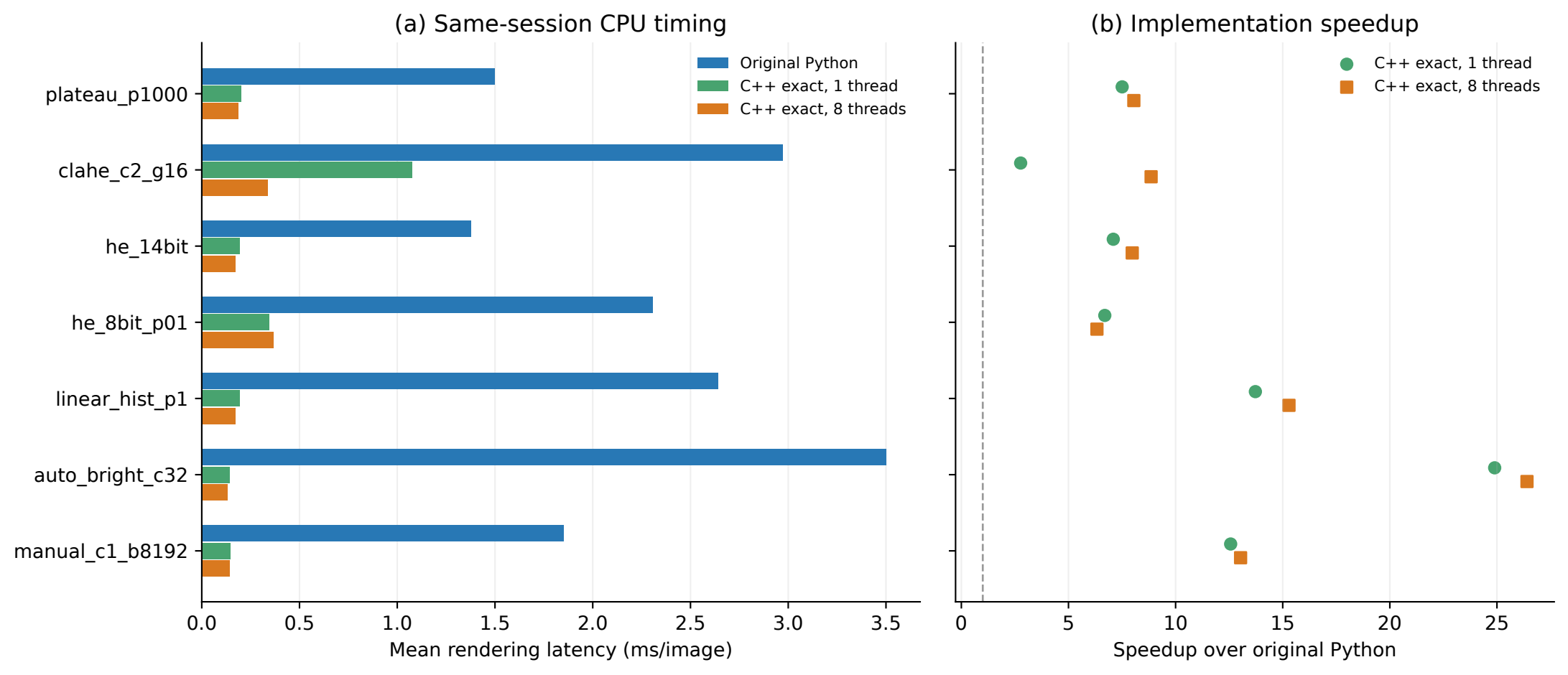
(a) Same-session CPU timing
Original Python
C++ exact, 1 thread
C++ exact, 8 threads
plateau_p1000
clahe_c2_g16
he_14bit
he_8bit_p01
linear_hist_p1
auto_bright_c32
manual_c1_b8192
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Mean rendering latency (ms/image)
(b) Implementation speedup
C++ exact, 1 thread
C++ exact, 8 threads
0
5
10
15
20
25
Speedup over original Python